

25PC2RA202: Sensors and Actuators Laboratory (S&A Lab)

Experiment 4: Shutter and Motion Artifacts

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Student Name: _____

Number: _____

1. Introduction

In this experiment, you will compare image sequences of a fan captured with a rolling shutter camera (Raspberry Pi Camera Module 3) and global shutter camera (Raspberry Pi Global Shutter Camera) on a Raspberry Pi 5. You will identify visible geometric distortion and motion blur. You should be working in teams of 4.

2. Hardware

- Raspberry Pi 5 (4 GB) with official 27 W USB-C power supply.
- Raspberry Pi Camera Module 3 - Rolling Shutter Camera.
- Raspberry Pi Global Shutter Camera with lens.
- Preferably, use the **ceiling fan** for your experiments. You can also use a standing **fan** if it is available in the lab. Alternatively, you could potentially bring your own **portable mini fans** ([Example](#)).

Evaluation Rubric

Assessment Component	Marks	Awarded Marks
Wiring	1	
Programming	2	
Observation Table	5	
Report Quality	1	
Attendance	1	
Total	10	

3. Instructions

a. Preparation:

- i. Shut down the Raspberry Pi and disconnect USB-C power before connecting a camera.
- ii. Connect Camera Module 3 first (if it is not already connected). Boot the Pi and run (in terminal) `rpigam-hello --list-cameras` to verify cameras are attached properly and corresponding camera number.
- iii. Mount the camera rigidly. Use a plain background and bright, continuous light. One of the group members will have to hold the camera during the experiment.
- iv. Place the target fan at the working distance. Camera Module 3 can autofocus; focus the Global Shutter Camera manually at the lens.

b. Rolling Shutter Camera:

- i. Capture the fan while off, at low speed, and at high speed. The python code automatically saves an image sequence. Email/WhatsApp yourself the image sequence. You can select one best representative frame for your record.

c. Global Shutter Camera:

- i. Shut down the Pi and disconnect power before changing cameras. Connect the Global Shutter Camera, reboot, and verify its camera number.
- ii. Manually focus its lens. Keep the target distance, illumination, and framing as close as possible to the rolling-shutter setup.
- iii. Capture the fan while off, at low speed, and at high speed. The python code automatically saves an image sequence. Email/WhatsApp yourself the image sequence. You can select one best representative frame for your record.

d. Saving, Printing, Reporting:

- i. In your record, print the paired images into the record notebook. Do not print every captured frame; print ONLY the selected representative pairs. Place the rolling- and global-shutter frames side by side at the same displayed size. Label the camera and condition.
- ii. Complete observation table below.

e. Rolling Shutter vs Global Shutter - Key Difference:

- i. Rolling shutter: image rows are recorded sequentially, so fast motion can make fan blades appear bent, skewed, stretched, or unevenly spaced.
- ii. Global shutter: the whole frame is exposed at essentially the same instant, so blade geometry and relative spacing are preserved.
- iii. Motion blur: either camera can still show softened or smeared edges when the fan moves during the exposure; blur alone is not a rolling-shutter artifact.

4. Observation Table:

Expected Observations: When the fan is rotating, the rolling-shutter image is expected to show a change in apparent blade geometry, such as bending, skew, stretching, or uneven spacing, because different image rows are captured at different times. The global-shutter image should preserve the fan's overall blade shape and relative spacing. Either image may still show motion blur: motion blur softens or smears the blade edges, whereas a rolling-shutter artifact changes the apparent shape or position of the blades across the frame.

Scene	Motion	Rolling Shutter	Global Shutter	Differences
Fan	Off			
Fan	Low Speed			
Fan	High Speed			



Figure 1 Representative rolling-shutter frame: the moving blades show geometric skew/warping, with some motion blur.

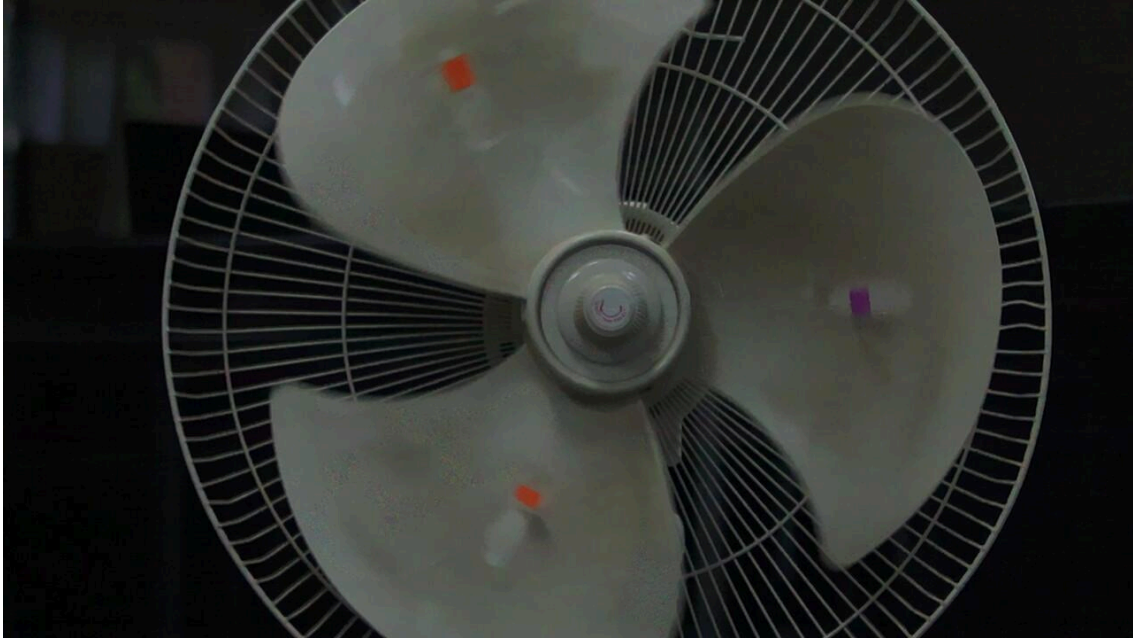


Figure 2 Representative global-shutter frame: blade geometry is preserved, although residual motion blur remains

5. Observations: Answer the following questions in your record:

- a. Do the blades appear straight or bent? Do all blades appear to have the same shape?
- b. Does the spacing between blades look uniform?
- c. Are global shutter blades sharp or blurred?
- d. Does the global shutter image show a simpler instantaneous shape?
- e. Which camera preserves the true geometry of moving objects more reliably, and can either camera still produce motion blur?

Signatures

Prof. A. Adithya Kashyap

Date:

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Date: